Rollout Gates and Privileged Distillation for Lane Selection with Sampled Safety: When Is Run-Time Simulation Enough?

Hossein Nick Zinat Matin* *Member, IEEE*, Jeffrey Sheng, Xiaoqian Gong, and Jesse Read

***Abstract*—A lane change can be decided by simulating its consequences or by a learned rule that predicts them; we study when each suffices, for a vehicle whose learned policies act through a sampled acceleration filter. We give the exact set of states from which a vehicle with held inputs can always keep its front clearance against a bounded-braking leader, with a nonempty least-intervention input interval, a sharp sampling penalty $b\Delta^2/8$ and a convex lane-commitment reserve whose terminal set is this kernel; a loss bound for a rollout gate that charges every executed decision, including those taken while a query is pending; and a split of its error into a price of unobserved information and a rule-specific term. Run in the loop in nominal traffic, the exact filter removed 70–90% of the runs that ended without an admissible input, raised every rule's return, and halved the privileged gate's margin over the learned rule with two lanes. In stochastic traffic with two and three lanes, the value of a prior-sampling query over a student distilled from a privileged gate rises with the uncertainty the traffic model describes, also with the policies held fixed, and changes sign; traffic information 10–50 ms old costs it at most 0.26, a full step of delay removes it. Its gap to a privileged query lies in the leaders' target generator, not their current state. No vehicle collided in 3.4 million runs, each ending at the horizon or when no admissible input remained.**

I. Introduction

Lane selection couples a discrete choice of traffic environment with continuous longitudinal control, and the choice can be made in two ways: a planner simulates the candidate change and compares it with staying, no worse than the base policy with an exact model, but its simulations must finish within the control period and are only samples under uncertain traffic; or a learned rule predicts the comparison in a fraction of a millisecond, with errors to be controlled. This paper asks, for a concrete hierarchy, *when run-time simulation is enough and when learning is needed*. Our answer: when the uncertainty that matters is the kind the declared model samples, the query fits within the control step, and its answer is used without waiting a step; otherwise, in our hierarchy, a student distilled from a privileged gate, or a cascade of the two, does better; under either, an exact sampled filter keeps the front clearance.

Related work. MOBIL derives lane-change incentives from car-following models [1]; responsibility-sensitive safety [2] and full-braking checks of car-following rules [3] and zero-order-hold braking analyses for cruise control [4] give formal envelopes, and reinforcement learning with analytic braking constraints has been applied to lane changes [5], [6]; our Section III differs in certifying the sampled, held-input closed loop exactly (intersample clearance, residual step, terminal viability) through its exact kernel, residual-step penalty and terminal set. Control barrier functions [7], [8] and predictive safety filters with stored tails [9] separate safety from learning; fail-safe trajectories verify motion plans online [10]. Rollout with several base heuristics is no worse than each [11], [12], with error terms of performance-difference form [13]; belief-conditioned simulation is the planning alternative to prior sampling [14], and inferring other drivers' internal states was shown to pay for freeway lane changes [15], and our prior-sampling query is one sampled planner, not a representative of all. Uniform coverage over sample prefixes is a confidence-sequence problem [16], a tool in Theorem 3, whose content is the charge accounting. On-policy relabelling [17] and privileged teachers [18] are studied in imitation learning, including the gap an observation-limited student cannot close [19].

Contributions. (i) *Exact sampled safety*: the robust front-viability kernel of the held-input longitudinal model, its nonempty least-intervention input interval, the sharp $b\Delta^2/8$ sampling penalty, and a lane-commitment reserve with an exact conic terminal set and certified front continuation, with an impossibility result for rear failures (Section III). (ii) *Performance accounting for query-assisted decisions*: a loss bound charged at flipped decisions, extended to asynchronous queries with sampling error, stale answers and decisions executed while waiting; a decision-error identity whose information price is rule-independent at a fixed observable history; and supporting sample, bias, compute and certification bounds (Section IV). (iii) *An empirical comparison under stated information and latency conditions* of a rollout gate, a student distilled from a privileged gate and their cascade, over two traffic models, two and three lanes, an uncertainty sweep, timing and delay, with controls, and the filters of (i) in the loop, which change the measured value of the decision layer (Section V). Theorems 1–3 and Corollary 1 are the main results; the others are supporting tools.

II. Longitudinal Hybrid Model

The ego vehicle has position x, speed v, and lane $q \in \mathcal{L} = \{1, \dots, L\}$, with $\dot{x} = v$, $\dot{v} = u$, $u \in [\underline{u}, \overline{u}]$, $0 \le v \le v_{\max}$. Neighbor maps select the immediate leader and follower of the ego's longitudinal position in each lane p, with speeds $v_p^{\pm}$

H. Nick Zinat Matin and J. Read are with the Laboratory of Computer Science (LIX), Ecole Polytechnique, Palaiseau, France. J. Sheng and X. Gong are with the Department of Mathematics, Amherst College, MA, USA. hossein.matin@polytechnique.edu, jsheng26@amherst.edu, xgong@amherst.edu, jesse.read@polytechnique.edu. *Corresponding author.

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and accelerations $a_p^{\pm}$, and bumper clearances $d_p^{\pm}$. A change to an adjacent lane $p \in \mathcal{N}(q)$ at t_k sets $q(t_k^+) = p$ and a clock $\eta(t_k^+) = 0$, and commits to a longitudinal option of duration $T_m = N\Delta$ during which the ego is counted in both lanes, so the old follower keeps reacting to it (*commitment model*). A stay option lasts Δ. Followers use the optimal velocity model (OVM) [20], $\dot{v}_i = \mathrm{clip}(\varrho_i(V_i(d_i) - v_i); [-b_f, a_f])$ with $V_i(d) = v_i^{\max} \frac{\tanh((d-d_c)/\ell_c)+\tanh(d_c/\ell_c)}{1+\tanh(d_c/\ell_c)}$ nondecreasing in the gap d; leaders are exogenous, followers react to the ego. The safety results need only valid bounds on neighbor accelerations over the certified interval,

$$a_p^+(t) \geq \underline{a}_p^+, \qquad a_p^-(t) \leq \overline{a}_p^-. \tag{1}$$

Deterministic traffic: leaders follow $\bar{v}_i + A_i \sin(\omega_i t + \phi_i)$, $\underline{a}^+ = -A_i|\omega_i|$, and follower bounds come from the monotonicity of the declared OVM: over a window of length T the gap to the ego is at most $d^{\uparrow} = d + (v - v_i)_+ T + \frac{1}{2}(\overline{u} + b_f)T^2$ for every admissible ego input sequence and the follower's speed is at least $v^{\downarrow} = v_i(1 - \varrho_i \Delta/n_s)^{n_s T/\Delta}$ (n_s integration substeps per sample, since $\dot{v}_i \geq -\varrho_i v_i$), so $\overline{a}^- = \mathrm{clip}(\varrho_i(V_i(d^{\uparrow}) - v^{\downarrow}); [-b_f, a_f])$ bounds its acceleration on the window, whatever the ego does. *Stochastic traffic:* leaders track a random piecewise-constant target speed with gain k_L, their acceleration clipped to $[-A_i|\omega_i|, A_i|\omega_i|]$; targets follow the sinusoid with noise and, with probability p_b per segment, a hard-braking target; follower gains ϱ_i and desired speeds are hidden from the controller, and $\overline{a}^-$ is the worst case over their declared ranges (the largest desired speed, and the gain range through seven values including its end points, where the unclipped bound, convex in the gain, is largest and the monotone clip keeps it largest). Bounds (1) hold for every realization. The nominal command u_c of a frozen longitudinal policy always passes through the filter below.

III. Longitudinal Safety and Lane Entry

For a required clearance $d_0 > 0$ let $H_p^{\pm} = d_p^{\pm} - d_0$, $D_p^+ = v_p^+ - v$, $D_p^- = v - v_p^-$, and write $\underline{u} = -b$ with $b > 0$ for the ego's braking authority. Barrier inequalities evaluated at sample times [8] do not certify the intersample behavior [21]; all certificates below hold in continuous time for inputs held over a step. For $H \geq 0$ and a window $T > 0$ let

$$m(H, D, T) = \inf_{0 < s \leq T} 2(H + Ds)/s^2, \tag{2}$$

which equals $-D^2/(2H)$ if $H > 0$, $D < 0$, $-2H/D \leq T$, is $-\infty$ if $H = 0$, $D < 0$, and $2(H + DT)/T^2$ otherwise, and let $I_p^T = [\ell_p^T, r_p^T]$ with $\ell_p^T = \max\{-b, -v/T, \overline{a}_p^- - m(H_p^-, D_p^-, T)\}$, $r_p^T = \min\{\overline{u}, (v_{\max} - v)/T, \underline{a}_p^+ + m(H_p^+, D_p^+, T)\}$.

Proposition 1 (Held-input certificate)**.** *On a window $[t, t+T]$ with constant neighbor maps and bounds* (1) *valid for every lane p in a set $\mathcal{P}$, $H_p^{\pm}(t) \geq 0$, and $0 \leq v(t) \leq v_{\max}$, any constant $u \in \bigcap_{p \in \mathcal{P}} I_p^T$ keeps $d_p^{\pm} \geq d_0$, $p \in \mathcal{P}$, and $0 \leq v \leq v_{\max}$ on the window: with $\mathcal{P} = \{q\}$, $T = \Delta$ every step of $u = \mathrm{clip}(u_c; I_q^{\Delta})$ is certified; with $\mathcal{P} = \{q, p\}$, $T = T_m$, one held input certifies a change in both lanes.*

Proof. Integrating the bounds twice, $H_p^+(t+s) \geq H_p^+(t) + D_p^+ s + \frac{1}{2}(\underline{a}_p^+ - u)s^2$ and $H_p^-(t+s) \geq H_p^-(t) + D_p^- s + \frac{1}{2}(u - \overline{a}_p^-)s^2$, nonnegative for $s \leq T$ by (2). □

This is the filter and entry guard of the experiments (the filter clips further onto a sample-time barrier interval when the two intersect, which only narrows I_q^{Δ}). It certifies one step or one commitment, not a continuation; the admissible set can empty. The rest of this section says exactly when it need not.

A. Exact front viability for sampled inputs

Consider one front neighbor with speed $v^+ \geq 0$ that does not reverse and brakes no harder than $b_L = -\underline{a}^+ \geq 0$, bounds and identity valid over the protected time; the ego's acceleration is held over each step, with $u \in [-b, \overline{u}]$ and $0 \leq v \leq v_{\max}$ enforced through the state-dependent bound $u \geq -v/\Delta$ (as in I_q^{Δ}), and state and actuation are exact. The emergency input is $u_{\mathrm{e}}(v) = -\min\{b, v/\Delta\}$; applied repeatedly it gives sample speeds $e_k(v) = (v - kb\Delta)_+$, interpolated affinely between samples, with displacement $E_{\Delta}(v, t)$. The leader's minimal displacement is $L(v^+, t) = \int_0^t (v^+ - b_L s)_+ ds$. Define

$$D_{\Delta}(v, v^+) = \max_{t \geq 0} \{E_{\Delta}(v, t) - L(v^+, t)\}, \tag{3}$$

attained on $[0, \Delta\lceil v/(b\Delta)\rceil]$, and the kernel $\mathcal{K}_{\Delta} = \{H^+ \geq D_{\Delta}(v, v^+)\}$. For a first input u, put $X_u(s) = vs + \frac{1}{2}us^2$, $v_u = v + u\Delta$, $v_{\Delta}^+ = (v^+ - b_L\Delta)_+$, and

$$\Psi_{\Delta}(u) = \max\Big\{ \max_{0 \leq s \leq \Delta} \{X_u(s) - L(v^+, s)\}, \\ X_u(\Delta) - L(v^+, \Delta) + D_{\Delta}(v_u, v_{\Delta}^+) \Big\}. \tag{4}$$

Theorem 1 (Exact sampled front kernel)**.** *Under the standing assumptions of this subsection, at a sample instant an input sequence keeping $H^+ \geq 0$ for all future time against every leader of the declared class (nonreversing, braking no harder than b_L; its extremal member brakes at b_L to rest) exists if and only if the state lies in $\mathcal{K}_{\Delta}$. For such a state, the first inputs that admit a safe continuation form the nonempty closed interval*

$$\mathcal{U}_F = \{u \in [u_{\mathrm{e}}(v), \min\{\overline{u}, (v_{\max} - v)/\Delta\}] : \ \Psi_{\Delta}(u) \leq H^+\}, \tag{5}$$

so projecting any nominal input onto $\mathcal{U}_F$ keeps the state in $\mathcal{K}_{\Delta}$ at every sample and the clearance between samples. D_{Δ} is continuous, convex and nondecreasing in v, nonincreasing in v^+, and computable from at most seven candidate times.

Proof. Every admissible ego sequence satisfies $v_{k+1} \geq (v_k - b\Delta)_+$, so its speed and displacement dominate the sampled braking envelope e_{Δ} and its integral E_{Δ}; every admissible leader dominates L, which maximal braking to rest attains. Hence u_{e} is safe on $\mathcal{K}_{\Delta}$, and if $H^+ < D_{\Delta}$ that leader defeats every ego policy. With the first input fixed, emergency braking afterwards minimizes displacement; splitting at Δ with $L(v^+, \Delta + s) = L(v^+, \Delta) + L(v_{\Delta}^+, s)$ gives (4) exactly, with $\Psi_{\Delta}(u_{\mathrm{e}}(v)) = D_{\Delta}(v, v^+)$; an actual leader ends the step with no smaller gap and speed, so membership is preserved. $E_{\Delta}(v, t)$

is a nonnegative combination of the convex $(v - kb\Delta)_+$, so Ψ_Δ is convex and nondecreasing in u and its sublevel set containing $u_{\mathrm{e}}(v)$ is an interval; on the stopping interval $E_\Delta - L$ is piecewise quadratic, whose breakpoints and zeros of relative speed are the candidates. □

The prefix in (4) cannot be dropped ($b = 6$, $\Delta = 0.1$, $v = 0.6$, $v^+ = 0.3$, $b_L = 0$, $H^+ = 0.008$: the terminal test alone accepts $u = -5$, whose prefix deficit reaches 0.009). Fronts protected only until T use the maximum over $[0, T]$, and several fronts are certified jointly, one inequality each.

Corollary 1 (Sharp sampling penalty)**.** *Let D_0 be* (3) *with continuous braking, $E_0(v,t) = \int_0^t (v - bs)_+ ds$, and $r = v - b\Delta\lfloor v/(b\Delta)\rfloor$. Then $0 \le D_\Delta - D_0 \le \frac{r}{2}(\Delta - r/b) \le b\Delta^2/8$, and the last constant is sharp.*

Proof. The two emergency trajectories coincide until the residual step, after which their displacement difference is nonnegative, nondecreasing, and ends at $r\Delta/2 - r^2/(2b)$; maximize over r. A stationary leader and $r = b\Delta/2$ attain the bounds. □

Sampling thus costs at most $b\Delta^2/8$ of clearance (7.5 mm at 10 Hz with $b = 6$) in the model that keeps $v \ge 0$ through the held-input constraint $u \ge -v/\Delta$ (a vehicle that stops and rests within a sample has a different residual step), and a continuous-time margin ε suffices for sampled viability whenever $\Delta \le \sqrt{8\varepsilon/b}$; $\mathcal{K}_\Delta$ is an invariant of sample instants, and an off-grid state with remaining hold h is certified by (4) with prefix h.

B. Terminal-safe lane commitments

The experiments admit a change only if the entry guard $I_q^{T_m} \cap I_p^{T_m} \neq \emptyset$ holds and hold that input; this certifies the commitment but not what follows it, and cannot react to the traffic. Let $T_m = N\Delta$, $U = (u_0, \dots, u_{N-1})$, $v_{i+1} = v_i + u_i\Delta$, $X_{i+1} = X_i + v_i\Delta + \frac{1}{2}u_i\Delta^2$ with $v_0 = v$, $X_0 = 0$, and $X_U(i\Delta + s) = X_i + v_i s + \frac{1}{2}u_i s^2$. For each occupied lane $p \in \{q, p'\}$ with rear speed v_p^- and bound $\overline{a}_p^-$, let $R_p(t) = v_p^- t + \frac{1}{2}\overline{a}_p^- t^2$. The plan set $\mathcal{F}_N(y)$ is defined by the input and endpoint speed bounds, by

$$H_p^+ + L_p(v_p^+, t) - X_U(t) \ge 0, \quad H_p^- + X_U(t) - R_p(t) \ge 0, \tag{6}$$

for $0 \le t \le T_m$ and both lanes, and by the terminal condition in the target lane, $H_F \ge D_\Delta(v_N, v_F^+)$ with $H_F = H_{p'}^+ + L_{p'}(v_{p'}^+, T_m) - X_N$ and $v_F^+ = (v_{p'}^+ - b_L T_m)_+$, the bounds being fixed at entry and valid through the commitment and, for the target front, afterwards.

Theorem 2 (Conic reserve and terminal handoff)**.** *$\mathcal{F}_N(y)$ is compact, convex, and exactly representable by finitely many second-order-cone constraints; when nonempty, its first-input projection is a closed interval. Executing the first input of a feasible plan and replanning in $\mathcal{F}_{N-1}$ at the next sample is feasible until the commitment ends, the stored tail being a fallback; both lanes' clearances are kept throughout, and the target-lane state belongs to $\mathcal{K}_\Delta$ at completion (front, not rear, continuation; Proposition 2).*

Proof. Positions and speeds are affine in U, so (6), split at sample and neighbor stopping times, is a family of quadratics in t with coefficients affine in U. For fixed t, $E_\Delta(\cdot, t)$ is convex and affine on each speed band $[jb\Delta, (j+1)b\Delta]$, hence the maximum of its $\lceil v_{\max}/(b\Delta)\rceil$ affine band extensions E_j, and the terminal condition becomes $H_F + L(v_F^+, t) - E_j(v_N, t) \ge 0$ for all j, $t \le (j+1)\Delta$, quadratics of the same kind. A quadratic $Az^2 + Bz + C$ is nonnegative on $[0,1]$ iff some $\lambda \ge 0$ makes $\begin{bmatrix} C & (B-\lambda)/2 \\ (B-\lambda)/2 & A+\lambda \end{bmatrix} \succeq 0$ (add $\lambda z(1-z)$; conversely $\lambda = [B + 2C]_+$ works), a rotated second-order cone with one auxiliary variable. After one step the actual neighbors lie on the safe side of their envelopes, which are monotone in their initial state and time-consistent, so the old tail stays feasible; at completion the front gap and leader speed can only have improved, and D_Δ is nonincreasing in the latter. □

This is a stored-plan safety filter [9] with an exact terminal set; minimizing $(u_0 - u_c)^2$ over $\mathcal{F}_N$ gives the least intervention with a compatible tail. Within $\mathcal{F}_N$ the two-input plans $u_1 = \cdots = u_{N-1}$ and the constant plans $u_0 = \cdots = u_{N-1}$ form nested convex restrictions $\mathcal{F}_{\mathrm{c}} \subseteq \mathcal{F}_2 \subseteq \mathcal{F}_N$; the experiments' guard $I_q^{T_m} \cap I_p^{T_m}$ is a different object, a constant plan without the terminal condition, and is incomparable with $\mathcal{F}_{\mathrm{c}}$ (at $v = 20$, $H^+ = 18$ against a stationary leader it admits $u = -6$ although $D_\Delta(20, 0) = 33.3$). The inclusions can be strict (examples in [22]). A follower keeping its own sampled stopping distance (Theorem 1 on its side) would make the joint continuation recursively feasible; ours are unfiltered OVM vehicles, and we claim nothing of the kind.

Proposition 2 (Rear obstruction)**.** *With constant neighbor maps and bounds* (1)*, if the follower may accelerate as hard as the ego, $\overline{a}^- \ge \overline{u}$, and is closing, $D^- < 0$, then for every input the worst-case rear clearance satisfies $H^-(t+s) \le H^-(t) + D^- s$ (integrate $u - a^- \le 0$ twice), so no input sequence certifies the rear constraint beyond $s = H^-(t)/|D^-|$; this requires an adversary that can sustain $\overline{a}^-$, which a bound alone does not establish for the OVM.*

The sweep uses Proposition 1 as filter and guard; Section V-E runs (5) and the restrictions of $\mathcal{F}_N$ against it. Theorems 1 and 2 say which failures a filter can avoid: every front failure from a state in $\mathcal{K}_\Delta$ and every failure during a commitment admitted by $\mathcal{F}_N$; rear failures between commitments against a closing follower that accelerates as hard as the ego are not.

IV. Rollout Gate: Decisions, Errors, and Compute

A. Hierarchy

The options s and c_p last Δ and T_m; s is admissible iff $I_q^\Delta \neq \emptyset$, and c_p iff the entry guard holds, $I_p^\Delta \neq \emptyset$ with a closing-rate margin in lane p, $\eta \ge \tau_d$, and $t + T_m \le T_f$. The option reward is $R_k = \int (w_d v - w_u u^2)\, d\tau - c_0 \mathbf{1}_{\{a_k \neq \mathsf{s}\}}$, with a failure charge C_{fail} when no action is admissible; values are undiscounted to T_f. The longitudinal policy is trained by PPO [23] through the filter and the lane-selection critic Q_ψ by Double DQN [24] with a validated training-time switching

cost $\bar{c}_0 \geq c_0$. Double DQN suits this small discrete choice made once per option, with replay of the rare changes and less overestimation favoring them; the bounds below hold for any proposal rule π_b. Here y is the full state (positions, speeds, lanes, clocks, hidden follower parameters, and the leaders' current targets, segment ends and generator parameters, all used by the privileged gate) and $o = \phi(y)$ the observation (Section V-A); the critic is deployed as $\pi_b(y) = \arg\max_{a \in \mathcal{A}(y)} Q_\psi(o, a)$, ties to s. Values V_b, Q_b, V_s are functions of y; $G(\pi)$ is the realized return on a scenario.

B. The gate and its loss

Let π_s stay whenever staying is admissible and act as π_b otherwise. At a *proposal epoch* k, $\pi_b(y_k) = \mathsf{c}_p$ and s is admissible; the advantage of the proposal over never changing is

$$g_k = Q_b(y_k, \mathsf{c}_p) - V_s(y_k). \tag{7}$$

The *privileged gate* executes the change iff $g_k \geq 0$; every deployable rule estimates the sign of g_k.

Proposition 3 (Loss of a gate)**.** *Let π follow π_b except at proposal epochs, where it decides from the past and from private randomness ξ independent of the transition noise, and let $\mathcal{E}$ be the epochs at which it decides differently from $g_k \geq 0$. For a Markov declared model*

$$V^\pi(y_0) \geq \max\{V_b(y_0), V_s(y_0)\} - \mathbb{E}^\pi \textstyle\sum_{k \in \mathcal{E}} |g_k|, \tag{8}$$

pathwise for every ξ if the model is deterministic (proof: the telescoping step of Theorem 3 with the true charges).

With exact advantages the gate is never worse than either base policy [11]; (8) is the performance-difference form [13]. A deployable gate decides from intervals rather than from g_k; the next result charges every executed decision, including those taken while a query is pending.

Theorem 3 (Charged asynchronous switching)**.** *Let h be the complete observable history (observations, actions, clocks, masks, controller state), $B(h)$, $S(h)$ the expected remaining returns of π_b, π_s given h, $a(h) = B(h) - S(h)$, and let returns lie in an interval of length $\mathcal{R}$. A query j dispatched at h_j returns i.i.d. paired rollouts with mean μ_j and sub-Gaussian proxy σ_j^2, with randomness independent of the traffic given h_j; suppose certified allowances $|\mu_j - a(h_j)| \leq \beta_j$ and $|a(h_k) - a(h_j)| \leq D_{j,k}$, and risks δ_j with $\sum_j \delta_j \leq \delta$, fixed before the query's samples are drawn. With $\bar{Z}_{j,n}$ the mean of the first n rollouts of query j and $r_j(n) = \sigma_j \sqrt{(2/n)\log(\pi^2 n^2/(3\delta_j))}$, the intervals $I_{j,k,n} = \bar{Z}_{j,n} \pm (r_j(n) + \beta_j + D_{j,k})$, intersected with $[-\mathcal{R}, \mathcal{R}]$, contain $a(h_k)$ at every executed proposal simultaneously with probability at least $1 - \delta$. Let a gate choose at every proposal epoch k, with current interval $[L_k, U_k]$ (the intersection of the intervals received so far, $[-\mathcal{R}, \mathcal{R}]$ while none has, and the last nonempty one if it empties), an action $d_k \in \{b, s\}$ (change, stay) and log $q_k(b) = [-L_k]_+$, $q_k(s) = [U_k]_+$. Then*

$$V^\pi(h_0) \geq \max\{B(h_0), S(h_0)\} - \mathbb{E}_\pi \textstyle\sum_{k \in \mathcal{P}^\pi} q_k(d_k) - \mathcal{R}\delta, \tag{9}$$

where $\mathcal{P}^\pi$ contains every proposal on the trajectory, answered or not.

Proof. Coverage is a union bound over prefixes ($\sum_n 6\delta_j/(\pi^2 n^2) = \delta_j$) and over adaptively dispatched queries; query randomness adds no information on the hidden state given h (the action likelihoods cancel in Bayes' rule), so the base Bellman equations hold in the gate's filtration. Policy switching [12] gives $W(h) \leq \mathbb{E}[R + W(h') \mid h, d] + c(h, d)$ with $W = \max\{B, S\}$, $c(h, b) = [-a(h)]_+$, $c(h, s) = [a(h)]_+$ and no charge outside proposals; telescoping over the finite horizon bounds the loss by the expected true charges. Couple π with the gate that follows it until the first invalid interval and then takes the sign of $a(h)$ (nonanticipative, since a is a function of h): before that time the true charges are at most the logged q_k, after it they vanish, the two gates differ with probability at most δ, and their returns by at most $\mathcal{R}$. □

A one-step delay thus enters (9) as the charge of the stay decisions taken while waiting, which a returned answer does not refund; (8) is the exact-charge special case. The stochastic experiments do not instantiate this certificate, because no certified β_j or $D_{j,k}$ is available there (the advantage can change by order one within a step); Section V reports the full-state decision charges as diagnostics instead.

C. Decision errors of learned and sampled rules

Lemma 1 (Decision-error identity)**.** *At a proposal epoch let z be any random variable (the observation o, the observable history h, or the full state y), $\bar{g}(z) = \mathbb{E}[g \mid z]$ and $\chi(z) = \frac{1}{2}(\mathbb{E}[|g| \mid z] - |\bar{g}(z)|)$. For every decision d that is conditionally independent of g given z,*

$$\mathbb{E}\big[|g|\, \mathbf{1}\{d \text{ wrong}\} \mid z\big] = \chi(z) + |\bar{g}(z)|\, \Pr\big[d \neq d^\star(z) \mid z\big],$$

where $d^\star(z)$ accepts iff $\bar{g}(z) \geq 0$ (ties broken either way). If $z' = \varphi(z)$ is a coarser statistic, $\mathbb{E}[\chi(z')] \geq \mathbb{E}[\chi(z)]$.

Proof. $\mathbb{E}[[\mp g]_+ \mid z] = \frac{1}{2}(\mathbb{E}[|g| \mid z] \mp \bar{g})$; the smaller is χ, the larger $\chi + |\bar{g}|$, incurred exactly when $d \neq d^\star$, an event independent of g given z. The last claim is Jensen's inequality for the concave $\min$ in $\chi(z) = \min\{\mathbb{E}[[-g]_+ \mid z], \mathbb{E}[[g]_+ \mid z]\}$. □

With $z = y$, $\chi = 0$: the privileged gate pays only its sampling error. With $z = h$, the observable history, the hidden quantities are exogenous and every deployable rule is a function of h, so their posterior given h, hence $\chi(h)$ and $\bar{g}(h)$, does not depend on the rule (its action probabilities cancel in Bayes' rule): at a fixed history, $\chi(h)$ is the price of what no observation-based rule can know, a rule using a statistic $o = \varphi(h)$ pays at least as much, and rules differ in the second term. The visited histories depend on the rule, so χ is the value of privileged selection at one decision, not a bound on a rule's return loss.

Proposition 4 (Learned, sampled and prior-sampled rules)**.** *In Lemma 1 with $z = h$: (i) a learned rule that accepts iff $\hat{g}(h) \geq 0$ has rule term at most $|\hat{g}(h) - \bar{g}(h)|$, hence at most the regression's mean absolute error on the visited proposals*

in expectation. (ii) A query averaging M rollouts that, given h, are i.i.d. with mean $\mu(h)$ and sub-Gaussian with variance proxy $\sigma^2(h)$ that accepts iff the sample mean is nonnegative has rule term at most $|\bar{g}|e^{-M\mu^2/(2\sigma^2)}$ if $\operatorname{sign}\mu = \operatorname{sign}\bar{g}$, hence at most $\sigma/\sqrt{eM}$ if $\mu = \bar{g}$, and at least $|\bar{g}|(1 - e^{-M\mu^2/(2\sigma^2)})$ otherwise. (iii) For every decision based on M i.i.d. rollouts $\mathcal{N}(\mu, \sigma^2)$ and every $\varepsilon > 0$, the larger of the rule terms at $\mu = \bar{g} = \pm\varepsilon$ is at least $\varepsilon\,\Phi(-\varepsilon\sqrt{M}/\sigma)$, attained by the sign of the sample mean.

Proof. (i) A wrong side needs $|\hat{g} - \bar{g}| \geq |\bar{g}|$. (ii) Chernoff, and $\max_x xe^{-Mx^2/(2\sigma^2)} = \sigma/\sqrt{eM}$. (iii) Two-point Bayes error under the uniform prior, attained by the sign of the sample mean. □

Proposition 5 (Bias of a prior-sampled query)**.** *Let $g(h, \theta) = \mathbb{E}[g \mid h, \theta]$, let p be the declared prior of θ and $p(\cdot \mid h)$ its posterior. A query that draws $\theta \sim p$ has sample mean $\mu(h) = \mathbb{E}_p[g(h, \theta)]$, and*

$$|\mu(h) - \bar{g}(h)| \leq 2\,\|g(h, \cdot)\|_\infty\,\mathrm{TV}\big(p, p(\cdot \mid h)\big).$$

Under the assumptions of Proposition 4(ii), as $M \to \infty$ its rule term converges, when $\mu(h) \neq 0$, to $|\bar{g}(h)|\,\mathbf{1}\{\operatorname{sign}\mu(h) \neq \operatorname{sign}\bar{g}(h)\}$, whereas that of a consistent learned rule converges to zero. A query drawing θ from any history-dependent law q_h (such as ours, centred on the current speed) has mean $\int g(h, \theta)\,q_h(d\theta)$ and the same bound with $\mathrm{TV}(q_h, p(\cdot \mid h))$.

Proof. $\mu - \bar{g} = \int g(h, \theta)\,(p - p(\cdot \mid h))(d\theta)$; the limits are the law of large numbers for the sign of the sample mean (when $\mu = 0$ and the rollouts have positive variance, the central limit theorem gives $|\bar{g}|/2$), and the learned rule's by consistency. □

The study tests this dichotomy: where the observation informs θ, the prior-sampled query has a bias no number of rollouts removes; where the declared randomness dominates, its error is variance, which M removes.

Corollary 2 (Compute of a sampled decision)**.** *Consider a query that simulates A option sequences ($A = B_o^\ell + 1$ for B_o options at depth ℓ; $A = 2$ for the one-step gate) over a horizon T_h with M rollouts, and let one rollout with n_v vehicles cost $(T_h/\Delta)(\kappa_0 + \kappa_1 n_\mathrm{v})$ seconds. Keeping the worst-case rule term of Proposition 4(iii) (the Gaussian two-point margin at means $\pm\varepsilon$) at most $\varepsilon/4$ requires exactly $M = \lceil z_{3/4}^2\sigma^2/\varepsilon^2\rceil$ rollouts ($z_{3/4}^2 \approx 0.455$), hence computation*

$$\mathcal{W} \approx 0.455\,\frac{\sigma^2}{\varepsilon^2}\,A\,\frac{T_h}{\Delta}\,(\kappa_0 + \kappa_1 n_\mathrm{v})$$

(up to the rounding of M), linear in n_v; a one-step deadline on P cores needs $\mathcal{W} \leq P\Delta$ and, for equal rollouts of cost c ideally scheduled, $c\lceil AM/P\rceil \leq \Delta$.

This is a yardstick for this design, not a lower bound over all queries; a faster simulator changes κ_0, κ_1, not σ^2/ε^2.

D. Certificates for the deployed rule

A conformal band on per-proposal residuals is not valid at deployment, because the student's decisions change the visited proposals (Section V-E reports its coverage); we relabel the proposals the student visits [17] and certify whole scenarios, which are independent, by Learn-then-Test [25].

Corollary 3 (Episode-loss requirement)**.** *Fix candidates $\pi_1, \ldots, \pi_J$ independently of n i.i.d. calibration scenarios on which all policies see the same random traffic, let $L^j = [\max\{G(\pi_b), G(\pi_s)\} - G(\pi_j)]_+$, k_j the number of scenarios with $L^j > \varepsilon$, and $U_j = \mathrm{Beta}^{-1}(1 - \gamma/J; k_j + 1, n - k_j)$ ($U_j = 1$ if $k_j = n$). With probability at least $1 - \gamma$, every candidate with $U_j \leq \alpha$ satisfies $\Pr[L^j > \varepsilon] \leq \alpha$ under this coupling (one-sided Clopper–Pearson bounds with a Bonferroni correction).*

V. Study

The code and a public companion [22] give all setups, parameter values, protocols, per-seed certification counts and per-experiment tables.

A. Setup

Model. $L = 2$ and 3 lanes with one leader and one follower each; $\Delta = 0.1\,\mathrm{s}$, $T_m = 1\,\mathrm{s}$, $T_f = 15\,\mathrm{s}$, $\tau_d = 2\,\mathrm{s}$, vehicle length $5\,\mathrm{m}$, $d_0 = 2\,\mathrm{m}$, $u \in [-6, 2.5]\,\mathrm{m/s^2}$, $v_{\max} = 20\,\mathrm{m/s}$; OVM followers with gain $\varrho = 0.8\,\mathrm{s}^{-1}$ (deterministic) or hidden $\varrho \sim \mathcal{U}(0.5, 1.1)$ and hidden desired speed $\mathcal{U}(15, 20)\,\mathrm{m/s}$ (stochastic); deterministic leaders with $A_i \leq 5\,\mathrm{m/s}$, $\omega \in [0.2, 0.8]\,\mathrm{rad/s}$, random phase; stochastic leaders track piecewise-constant targets with $k_L = 1\,\mathrm{s}^{-1}$ over segments of $\mathcal{U}(0.5, 3)\,\mathrm{s}$, a hard-braking target being $\bar{v}_i - 1.5A_i - 2\,\mathrm{m/s}$. Neighbors are integrated in 10 substeps per sample with exact position updates; clearances are monitored at every substep. Scenarios come from seven families (random, equal speeds, faster target lane, fast follower, braking leader, squeeze, late change), initial states admissible in both lanes. The *uncertainty level* ν scales the leaders' randomness: target noise $0.6\nu A\,\mathcal{U}(-1, 1)$ and hard-braking probability $\min\{0.2\nu, 0.6\}$ per segment; $\nu = 1$ is the nominal traffic. Rewards: $w_d = w_u = 1$, $c_0 = 2$, $C_\mathrm{fail} = 200$, so one unit of return is one metre of progress, half a switching charge, or 0.5% of a failure.

Rules. The *critic* π_b; the *privileged gate* (16 rollout pairs from y, counted as trajectories in Table I); the *prior-sampling gate*, the deployable query that draws follower parameters and future leader targets from a heuristic law q_h built on the declared ranges (Proposition 5), infers the current target from the measured acceleration, uses common random numbers in both branches, and averages M rollouts; the *student*, which accepts iff $\hat{g} > \tau$, τ validated per seed; the *cascade*, which runs the student and queries the prior-sampling gate only when $|\hat{g}|$ lies in a validated band. In deterministic traffic the rollout is exact, and the gate also rejects a change whose rollout ends in a failure (a veto that Proposition 3 charges). A blocking gate whose query returns after k steps stays meanwhile (the cascade decides by the student and waits only on the proposals

it queries); its acceptance is executed only if π_b still proposes the change and it is admissible.

Learning. PPO: 500 iterations of 64 episodes, reused everywhere. Double DQN (2×128 tanh, 6400 episodes) at each ν; $\bar{c}_0$ validated per seed over $\{2, \dots, 20\}$ (deterministic), $\bar{c}_0 = 10$ with $\{5, 10, 14\}$ validated at $\nu = 1$ (stochastic). *Inputs*: the critic sees o (26 numbers: time left, speed, cooldown; per lane the clearances, relative speeds, declared bounds and a validity flag; edge flags); the student, the cascade and the prior-sampling gate see o^+, which adds the three leaders' measured accelerations and the proposal's direction; the privileged gate sees y. The student is a 2×128 tanh regression on o^+, so in stochastic traffic it estimates $\mathbb{E}[\mathrm{clip}(\bar{g}_4; \pm 60) \mid o^+]$, with $\bar{g}_4$ the mean of four privileged rollouts (11–15% clipped), on exact labels (4096 episodes plus two on-policy rounds of 2048) or on means of $K = 4$ privileged rollouts (2048 plus one round of 1024; $4\times$ in a budget study). Validation (300 scenarios), calibration ($n = 500$, $\alpha = \gamma = 0.1$) and test sets (7 families of 100 scenarios, common to all rules) are disjoint; 15 and 10 seeds in deterministic traffic, 10 per level and 20 at $\nu = 1$ in the sweep.

Metrics. (a) *Return gain* ΔG over the critic in the same seed, with a 95% t-interval over training seeds on a common test panel (variation across trained policies, not across new traffic; the only interval type we use; no multiplicity correction, so we interpret only effects consistent across levels and lane counts) and the number of seeds with $\Delta G > 0$, decomposed into progress net of effort, switching charges and failure charges. (b) *Recovered share* $\Delta G / \Delta G_{\mathrm{priv}}$ of the privileged gate's gain. (c) *Decision charge*: $\mathbb{E}[|g| \mathbf{1}\{d \text{ wrong}\} \mid y]$ of Lemma 1 with $z = y$ on the student's held-out proposals. (d) *Work*: rollouts per episode, cores per step. (e) *Safety*: collisions, minimum clearance, runs ending with no admissible input. The *value of simulation* is $\Delta G(\text{prior-sampling gate}) - \Delta G(\text{student})$, paired by seed.

B. Deterministic traffic, then the uncertainty sweep

With exact rollouts the gate improves the validated critic by 1.16 and 1.43 in every seed (15 and 10 seeds), the student keeps 75% and 29% of the gain without simulation, the cascade 74% and 49% with 0.5–0.7 simulated decisions per episode, and (8) held pathwise in 20 000 check episodes: with the exact model, simulation is enough whenever it fits into the step. MOBIL [1] through the same filter (tuned per seed; fixed at $\nu = 1$) returns 5.3 and 8.8 less than the critic (8.9 and 11.3 at $\nu = 1$, no seed better): the gains below are over a competent base policy. Table I and Fig. 1 vary the uncertainty level from $\nu = 0$, where only segment lengths and the follower parameters are random, to $\nu = 2$, where 40% of the leaders' segments are hard-braking ones; ν raises noise and braking frequency together. Across gates and levels, failure charges contribute -0.2 to $+0.3$ to a gain and switching charges $+0.1$ to $+0.6$; the rest is progress net of effort.

Reference and learning. The privileged gate gains 0.58–1.10 (two lanes) and 0.83–1.23 (three) in every seed. The student gains at every level ($+0.13$ to $+0.67$ with two lanes, $+0.18$ to $+0.34$ with three; 8–10 of 10 seeds, 15–18 of 20 at $\nu = 1$); its share of the privileged gain falls from 61% to 21% with two lanes and stays at 18–27% with three. Four times the labels change its gain by -0.01 ± 0.12, -0.03 ± 0.15.

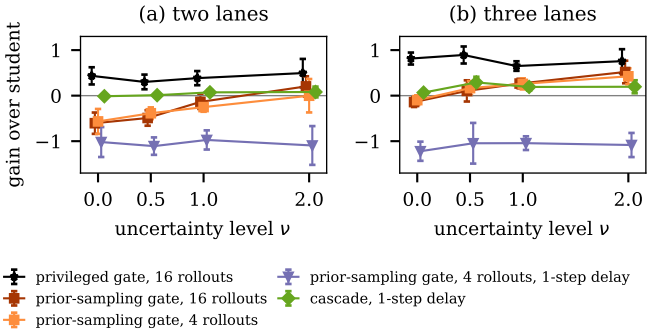


Fig. 1. Gain over the student by uncertainty level (95% t-intervals).

Simulation. At $\nu \leq 0.5$ with two lanes the prior-sampling gate does not beat the critic ($+0.06$ and -0.22 with 16 rollouts, no better than with 4): negative progress offset by fewer lane changes, the signature of the bias Proposition 5 allows. As the leaders become more random its gain grows to $+0.34$ (two lanes) and $+0.80$ (three) at $\nu = 2$, 54% and 77% of the privileged gain, with positive progress. The value of simulation rises with ν in both lane counts. With two lanes it is negative for $\nu \leq 0.5$ (-0.61 ± 0.23, -0.49 ± 0.17) and its point estimate changes sign between $\nu = 1$ (-0.14 ± 0.15) and $\nu = 2$ ($+0.20 \pm 0.22$). With three lanes it is negative at $\nu = 0$ (-0.14 ± 0.11) and positive for $\nu \geq 1$ ($+0.27 \pm 0.10$ in 18 of 20 seeds, $+0.52 \pm 0.25$ in 10 of 10). The critic is retrained at each level; with the $\nu = 1$ critic, student and gates frozen and only the traffic varied, the 16-rollout query's value over the student is -0.42, -0.47, -0.10, $+0.31 \pm 0.17$ (10 of 10 seeds) with two lanes and -0.01, $+0.12$, $+0.32$, $+0.52 \pm 0.22$ (10 of 10) with three: the rise belongs to the traffic, not to retraining; with two lanes the frozen policies also resolve the sign change, while with three the frozen value starts at -0.01 ± 0.19, so the negative value at $\nu = 0$ may partly reflect retraining.

What the privileged gate knows. We gave the prior-sampling gate (16 rollouts) one privileged item at a time, at $\nu \in \{0, 1, 2\}$ with 10 seeds. The true follower parameters change its gain by -0.15 to $+0.11$; the leaders' current target, segment end and target generator (the pattern's mean, amplitude, frequency, phase) together raise it by $+0.16$ to $+0.99$, to within 0.08 of the privileged gate. Separating the two, the generator alone raises it by $+0.13$ to $+0.95$, 81–96% of that gain (9–10 of 10 seeds at every level), and the current target and segment end alone by $+0.01$ to $+0.14$ (resolved in one of six cells); with the generator, simulation beats the student at every level ($+0.35$ to $+0.70$). Deployable queries that identify the follower parameters online, or condition the leaders on their observed history, gain nothing or lose up to 0.61. The sampler's gap is thus the leaders' pattern, which it redraws at every query, not their current state. This does not explain the student's wins: it can at best approximate $\mathbb{E}[g \mid o^+]$ (Lemma 1), so its labels

TABLE I

RETURN GAIN OVER THE CRITIC BY UNCERTAINTY LEVEL ν (95% t-INTERVAL OVER SEEDS; SEEDS POSITIVE; ROLLOUTS/EPISODE AT $\nu = 1$). CRITIC'S RETURN: 191.7 TO 177.5 (TWO LANES), 199.1 TO 188.0 (THREE). *ZERO DELAY; 10–50 MS OLD TRAFFIC COSTS IT 0.07–0.26. †SEES y.

	$\nu = 0$	$\nu = 0.5$	$\nu = 1$	$\nu = 2$	rollouts/ep.
Two lanes					
Privileged gate, 16 rollouts†	+1.10±0.61 (10/10)	+0.58±0.20 (10/10)	+0.63±0.15 (20/20)	+0.63±0.33 (10/10)	37
Prior-sampling gate, 16 rollouts*	+0.06±0.44 (4/10)	-0.22±0.19 (2/10)	+0.11±0.14 (12/20)	+0.34±0.19 (10/10)	41
Prior-sampling gate, 4 rollouts, 1-step delay	-0.35±0.53 (3/10)	-0.83±0.20 (0/10)	-0.73±0.22 (1/20)	-0.96±0.35 (1/10)	7
Student	+0.67±0.50 (10/10)	+0.28±0.13 (10/10)	+0.24±0.09 (18/20)	+0.13±0.11 (9/10)	0
Cascade, 1-step delay	+0.65±0.48 (10/10)	+0.29±0.12 (10/10)	+0.32±0.08 (20/20)	+0.22±0.20 (10/10)	1
Three lanes					
Privileged gate, 16 rollouts†	+1.00±0.21 (10/10)	+1.23±0.50 (10/10)	+0.83±0.12 (20/20)	+1.04±0.40 (10/10)	72
Prior-sampling gate, 16 rollouts*	+0.04±0.19 (8/10)	+0.44±0.52 (7/10)	+0.44±0.11 (19/20)	+0.80±0.38 (10/10)	72
Prior-sampling gate, 4 rollouts, 1-step delay	-1.04±0.32 (0/10)	-0.71±0.65 (1/10)	-0.86±0.18 (0/20)	-0.80±0.44 (2/10)	12
Student	+0.18±0.15 (9/10)	+0.34±0.37 (8/10)	+0.18±0.11 (15/20)	+0.28±0.22 (9/10)	0
Cascade, 1-step delay	+0.25±0.17 (9/10)	+0.63±0.46 (10/10)	+0.37±0.11 (18/20)	+0.48±0.29 (10/10)	3

buy a better observation-based rule, not the hidden state.

Delay and the cascade. A delay of one step costs the 4-rollout query 0.45–1.09 (two lanes) and 1.12–1.51 (three), paired by seed, so that it loses 0.35–1.04 to the critic at every level (Table I), all of it progress: the vehicle stays while waiting and changes later in a worse state. A critic delayed one step loses as much (0.41 vs. 0.44, one seed). The cascade, which waits only on the proposals it queries (0.1–0.7 per episode), gains $+0.07$ to $+0.29$ over the student with three lanes at every level and as much as the student with two. Over delays of up to 7 steps at $\nu = 1$ the 4-rollout gate's gain falls to -6.1 and -9.7, the cascade's stays within $+0.14$ to $+0.33$. To measure sensitivity to the age of the information (a stale-state study, not a demonstrated real-time pipeline), the proposal at the decision instant is evaluated from the traffic 10, 30 or 50 ms earlier and executed only if still admissible; with two lanes, where the direction is known in advance, an early query at every admissible epoch realizes this timeline. The lag costs the prior-sampling gates 0.05–0.26 (two lanes) and 0.07–0.18 (three), the privileged gate 0.01–0.11, against 0.45–1.51 for one step; with three lanes the 16-rollout gate still beats the student with 30 ms old traffic ($+0.25 \pm 0.15$, 10 of 10 seeds), and the cascade is unaffected.

Controls. A critic with a validated margin on its advantage gains -0.28 to $+0.02$ (the gates' gain is not caution); validating $\bar{c}_0 \in \{5, 10, 14\}$ changes it by $+0.28 \pm 0.33$, -0.27 ± 0.33. *Decision charge* (a diagnostic). Estimated with four privileged rollouts at the true state, it is biased upward by $(\mathbb{E}|\bar{g}_4| - |g|)/2$; on held-out proposals the student's charge is 0.16–0.45 (two lanes) and 0.40–0.61 (three), the prior-sampling gate's 0.9–3.0. An audit at $\nu = 1$ (references of up to 128 rollouts) changed the student's and gates' charges by 5–26%, enough to swap close pairs, and gave one privileged rollout a lower charge than the student in 18 of 20 seeds. On test proposals the student's charge is several times its held-out value and above the 4- and 16-rollout gates', yet its return is higher with two lanes: charges do not order rules by return.

C. Compute

The rollout with the filter of Proposition 1, compiled with Numba and timed on one core of an ARM laptop, follows Corollary 2 within 13% in all 42 timings, with $\kappa_0 \approx 4.7\,\mu$s and $\kappa_1 \approx 0.1\,\mu$s per vehicle and step. At $n_\mathrm{v} = 7$, $T_h = 15$ s one core runs 58 pairs of rollouts per 100 ms step: the 16-rollout one-step query takes 28 ms, the 4-rollout one 7 ms, and the 16-rollout two-step query over $B = 3$ options 1.4 cores. The spread σ of a privileged sampled advantage has root mean square 23.7 per proposal with two lanes (14.6 with three); as an illustrative input (not a worst case), Corollary 2 at $\varepsilon = 1$ asks $M = 256$, 4.4 cores, yet one privileged rollout usually beats the student's charge (Section V-B). At $n_\mathrm{v} = 7$ compute does not separate the student from a prior-sampling gate; the information in the rollouts does.

D. Information, labels, shift

A critic given o^+ (the student's inputs) gains $+0.21 \pm 0.25$ and $+0.49 \pm 0.74$; a student distilled on top of it adds $+0.31 \pm 0.32$ and $+0.13 \pm 0.25$, unresolved at 10 seeds. Under shifted test traffic (twice the hard-braking segments, noisier leaders) the student's gain shrinks to $+0.13 \pm 0.17$ and $+0.06 \pm 0.20$, the privileged gate's stays at $+0.71$, $+0.73$.

E. Certification and safety

With Corollary 3, $n = 500$ and $\alpha = \gamma = 0.1$, a candidate is certified iff $k_j \le 35$ ($J = 7$), separately for each tolerance $\varepsilon \in \{1, 2, 5\}$ fixed before calibration. At $\nu = 1$, among the student, the cascade, a router pricing the query's work at four levels and the prior-sampling gate with the measured latency of our Python simulator (up to 7 steps), with two lanes the student, the cascade and the router at a high price are certified for $\varepsilon = 1$ in 9–10 of 10 seeds ($k_j \approx 23$, $U_j \approx 0.07$), the delayed gate in none ($k_j = 160$); with three lanes $\varepsilon = 5$ is certified in 8–10 seeds ($k_j \approx 50$ at $\varepsilon = 1$). This bounds the probability of an episode loss above ε, a guarantee distinct from the filter's clearance and the return bounds of Section IV; per-proposal certification would be unreliable (a conformal band covered 0.78–0.83 against a nominal 0.90).

Safety. A run ends at T_f or when the filter has no admissible input (a failure). No vehicle collided in the 3.4 million evaluated runs of all experiments, and the minimum clearance was 2.000 m in both lanes of every commitment. 98.81–99.99% of

the critic's and gates' runs reached T_f; under the held-input filter 93% of the failures came through the front constraint.

Filters of Theorems 1 and 2. With the same frozen rules and scenarios at $\nu = 1$ we replaced the front constraint by the projection onto $\mathcal{U}_F$ in the current lane, and the entry guard by the existence of a terminal-safe constant ($\mathcal{F}_c$) or two-input ($\mathcal{F}_2$) plan, searched on grids of 41 and 19^2 plans, each checked exactly in continuous time and executed as stored. No initial state, and apart from one step of one run no state at which an input was applied, left $\mathcal{K}_\Delta$. The kernel filter cut the failures from 0.37–0.53% to 0.10–0.14% of the runs with two lanes and from 0.30% to 0.03% with three, and raised every rule's return by $+0.53$ to $+0.86$ (10 of 10 seeds) and $+0.62 \pm 0.44$ (6 of 10 seeds), all from the runs it saved: the kernel bound was active in 0.03–0.05 steps per run. Both terminal-safe guards gave identical results, rejected an entry the held-input guard admitted only 2–8 times per rule in 7000 runs, and removed the failures labelled front-only that remained under the kernel filter: each ended a held-input commitment in a state outside $\mathcal{K}_\Delta$, which the terminal condition of Theorem 2 excludes. What remains (7 and 2 of 7000 runs of the critic) are steps where the front kernel and a closing follower's rear constraint conflict; at these states no input satisfies both, and Proposition 2 is their rear-only analogue. The value of simulation over the student is unchanged (4 rollouts: -0.11 ± 0.12 and $+0.33 \pm 0.17$, against -0.19 and $+0.34$), but with two lanes half the privileged gate's margin over the student (0.25 of 0.45 on these 10 seeds) and all of the cascade's ($+0.09$) was failure avoidance the kernel now gives every rule.

VI. Discussion

What the uncertainty is about. The value of a prior-sampling query over the student rises with the uncertainty the declared model describes, and its sign changes between $\nu = 0$ and 1 with three lanes and between $\nu = 0.5$ and 2 with two; with the policies frozen the rise persists and the two-lane change is resolved by the traffic alone. The sampler's gap to the privileged reference lies in the leaders' target generator, their pattern, not in their current state or the follower parameters. Where the pattern matters, the student wins. Where the leaders' future is mostly the declared randomness, the sampler wins and recovers up to 77% of the privileged gain; a validated belief-conditioned sampler may close part of the rest.

When the answer arrives. Traffic information up to 50 ms old costs a gate little; a blocking gate waiting a whole step loses 0.45–1.51 of progress, charged by Theorem 3. By Corollary 2 the query stops fitting with more vehicles, a longer horizon, a tighter margin or a deeper lookahead; there the student or the cascade remains usable.

Safety and decisions. With two lanes half the privileged gate's margin over the student was failure avoidance that the kernel filter gives every rule: the safety layer should be exact before the decision layer is judged.

Limitations. The model is longitudinal with exogenous, non-anticipating leaders and reactive OVM followers, one pair per lane, and exact sensing; the conic reserve is run only in its constant and two-input restrictions; rear safety beyond a commitment needs a follower contract; and guarantees for learned components are statistical, relative to the declared model and calibration distribution.

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